

High-Voltage CAPACITOR DISCHARGER

T.K. HAREENDRAN

During repairs of vacuum tube equipment, switch-mode power supplies, microwave ovens, or similar systems, buffer or reservoir capacitors can retain potentially lethal voltages even after power is removed, requiring safe discharge. A high voltage capacitor discharge system provides controlled energy dissipation. Fig. 1 shows a typical high-voltage capacitor that must be discharged before servicing.

Some practitioners recommend short-circuiting charged capacitors with an insulated screwdriver. This method should be avoided. The resulting high discharge current can damage the capacitor and present a serious safety risk. Capacitors may also spark or fail violently. Fig. 2 highlights this hazard, as capacitors can retain dangerous high voltage for extended periods.

Fig. 3 illustrates a safer method

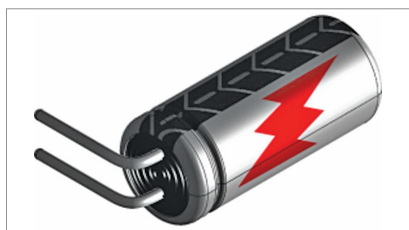


Fig. 1: A high-voltage capacitor



Fig. 2: Warning – capacitors can retain dangerous high voltage

for discharging a capacitor using a bleeder resistor. Passing the charge through a suitable power resistor limits current flow and reduces both component stress and operator risk.

The simple arrangement shown in Fig. 3 is suitable for capacitor voltages up to 600V

DC. A 4.7k Ω , 10W power resistor is generally adequate for this purpose. Initial tests used two 10k Ω ($\pm 5\%$), 5W ceramic-encased wire-wound resistors connected in parallel as the bleeder resistor (RD), and the setup performed reliably. Fig. 4 shows the resistors used for discharging.

One limitation of this basic method is the need to repeatedly check the capacitor voltage with a DC voltmeter to confirm that discharge has reached a safe level.

A more practical approach is a dedicated capacitor discharge probe system that keeps the circuitry simple while providing reliable, visible

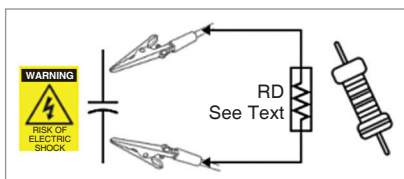


Fig. 3: Safe method to discharge a capacitor using a resistor

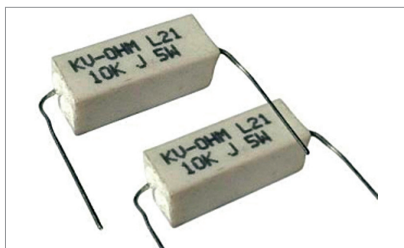


Fig. 4: Power resistors used as bleeder resistors

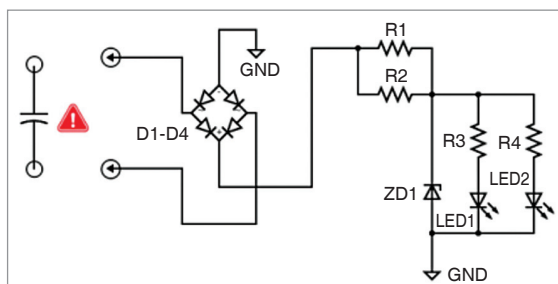


Fig. 5: Safe capacitor discharger circuit with bridge rectifier and indicators

operation. Fig. 6 shows the complete circuit with a bridge rectifier and LED indicators.

This passive discharger probe system enables safe and prompt discharge of high-voltage power-supply reservoir, filter, or buffer capacitors. The unpowered probe is placed across the capacitor terminals, allowing stored energy to dissipate at a controlled rate. The circuit has no polarity restriction and therefore operates correctly regardless of connection orientation.

As shown in Fig. 5, two 22k Ω /5W resistors (R1 and R2) are connected in parallel to form an 11k Ω bleeder resistor that drains the capacitor. Two LEDs (LED1 and LED2), each in series with a 330 Ω /¼W resistor (R3 and R4), provide visual indication.

PARTS LIST

Semiconductors:

- D1-D4 - 1N4007 rectifier diode
- ZD1 (1N4734A) - 5.6V, 1W Zener diode
- LED1, LED2 - 5mm LED amber/yellow

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

- R1, R2 - 22k Ω , 5W
- R3, R4 - 330 Ω
- NTC - 3 Ω

Capacitors:

- 150 μ F, 450V electrolytic (for charging)

Miscellaneous:

- Red, black - Alligator clips (two)
- Wires
- Capacitor to be discharged